Neural Style Transfer results of set id: 1

Content Image

Style Image

Stylized Image

Neural Style Transfer input parameters

Parameter	Value
Model	vgg19
Parameter set ID	1
Content Image	dancing.jpg
Style Image	warhol1.png
Stylized Image	stylized_1.jpg
Content layer	conv_4
Style layers	conv_1, conv_2, conv_3, conv_4, conv_5
Content weight	1
Style weight	1000000
Optimizer	LBFGS
Number of steps	100

Intermediate results visualization

Intermediate steps

step_50.jpg step_100.jpg

Neural Style Transfer stylized image assessment

Metric	Score	Interpretation
Perceptual Loss	51.114	
Content Loss	7.291	
Style Loss	43.823	
Perceptual metrics	0.41	
LPIPS	0.41	Somewhat different - clear differences.
Pixel-based evaluations	0.31	
SSIM	0.67	Low similarity. Content structure is partially lost.
PSNR	0.1	Bad pixel-level fidelity.
EMD	0.81	Moderate - clear differences.
Artifact detection	0.93	
High frequency noise	1.0	Excellent. Very low noise.
Unnatural edges	1.0	Excellent. Very few unnatural edges.
Color inconsistency	0.7	Acceptable. Moderately consistent.
Unique colors	1	Excellent color depth.
Coherence metrics	0.73	
Texture coherence	0.99	Extremely high - almost identical textures.
Edge coherence	0.56	Good edge coherence.
Aesthetic metrics	0.52	
Colorfulness	1.0	Extremely colorful - maximum saturation.
Contrast	0.33	Low contrast.
Sharpness	0.0	Blurry.
Composition	0.75	Exceptional - highly structured.

Overall score: 0.57

Metric descriptions

Perceptual loss metrics

Perceptual Content Loss

Perceptual Content Loss measures the dissimilarity between the stylized result and the original content image based on high-level feature representations extracted from a pre-trained VGG network. Unlike pixel-based metrics that compare individual pixel values, this metric evaluates differences in semantic content, object structures, and overall scene composition by analyzing how the images activate deep neural network layers trained to recognize visual concepts. It captures whether the essential 'what' of the image–the objects, shapes, and their spatial arrangement–remains intact after style transfer. Lower values indicate better preservation of content, while higher values indicate greater deviation from the original.

A lower value indicates that the stylized image successfully preserves the semantic content and structural essence of the original image. Objects remain recognizable, spatial relationships between elements are maintained, and the overall scene composition stays true to the content source. The style has been applied without compromising the fundamental information that allows viewers to identify what the image depicts.

Perceptual Style Loss

Perceptual Style Loss measures the dissimilarity between the stylized result and the style image based on texture, color, and pattern representations extracted from multiple layers of a pre-trained VGG network. This metric evaluates how well the artistic essence–brushstrokes, color palettes, textural qualities, and repetitive patterns–has been transferred from the reference style to the final image. It works by comparing the Gram matrix correlations of feature activations across different network layers, capturing stylistic characteristics rather than semantic content. Lower values indicate better style matching, while higher values indicate greater deviation from the intended style.

A low value indicates that the stylized image successfully captures the artistic essence of the reference style. The color palette, textural qualities, brushstroke patterns, and overall visual rhythm closely align with the style image, creating a cohesive artistic interpretation that feels true to the intended aesthetic. The style transfer has effectively extracted and applied the defining characteristics of the reference artwork.

Perceptual metrics

LPIPS

LPIPS measures the perceptual similarity between the stylized result and the original contentimage by comparing deep feature representations extracted from a pre-trained VGG network. Unlike traditional pixel-based metrics, LPIPS aligns more closely with human visual perception, as it evaluates differences in higher-level image structures and patterns rather than exact pixel values.

A high normalized value indicates that the stylized image preserves the perceptual essence of the original content very well. The image remains easily recognizable, with minimal perceptible differences in structure, texture, and overall composition.

A low normalized value suggests that the stylized image is perceptually distant from the content image. This may indicate excessive distortion, loss of recognizable features, or that the artistic style has overwhelmed the original content to the point where the subject becomes difficult to identify.

Pixel-based metrics

SSIM

SSIM ((Structural Similarity Index))measures the structural similarity between the stylized result and the original content image by evaluating three key components: luminance, contrast, and structural information. Unlike simple pixel-by-pixel comparisons, SSIM assesses how well the overall structures, patterns, and spatial relationships from the content image are preserved in the stylized output. This metric is particularly sensitive to distortions that affect human perception of image quality.

A high normalized value indicates that the stylized image maintains strong structural fidelity to the original content. Edges, shapes, object boundaries, and spatial arrangements remain clearly recognizable.

A low normalized value suggests significant structural degradation has occurred during the style transfer process. The stylized image may exhibit distorted object shapes, blurred boundaries, collapsed details, or unnatural spatial arrangements that compromise the recognizability of the original content.

PSNR

PSNR measures pixel-level fidelity between the stylized result and the original content image by calculating the ratio between the maximum possible pixel value and the mean squared error between the two images. It provides a quantitative assessment of how closely the stylized image matches the content image at the individual pixel level, reflecting the presence of noise, distortion, or exactness of reproduction.

A high normalized value indicates excellent pixel-level preservation, meaning the stylized image maintains precise color values, brightness levels, and fine details with minimal deviation from the original content.

low normalized value reveals substantial pixel-level discrepancies between the stylized result and the content image. This may manifest as visible noise, color distortion, loss of fine details, or general degradation in image fidelity.

EMD

EMD (Earth Mover's Distance) measures the similarity between the color distribution of the stylized result and that of the style image. It calculates the minimum amount of 'work' required to transform one color histogram into another, effectively quantifying how well the stylized image has captured the color palette, tonal range, and overall chromatic characteristics of the reference style image.

A high normalized value indicates that the stylized image successfully replicates the color characteristics of the style image. The color palette, saturation levels, and tonal distribution closely match the reference, suggesting effective transfer of the style's chromatic essence while maintaining natural color transitions.

A low normalized value reveals that the stylized image has failed to capture the color characteristics of the style image. The color distribution may differ significantly, manifesting as incorrect hues, mismatched saturation levels, or unnatural color transitions. This may result from inadequate style transfer optimization or interference from the content image's original colors overwhelming the intended style palette.

Artifact detection metrics

High Frequency Noise

High-Frequency Noise measures the presence of unwanted fine-grained artifacts and random pixel-level variations in the stylized image. It detects rapid intensity fluctuations across neighboring pixels that typically manifest as graininess, speckling, or texture irregularities not present in either the original content or intended style. Such noise often arises from optimization instabilities or over-exaggeration of fine style details during the transfer process.

A high normalized value indicates that the stylized image exhibits minimal high-frequency noise, appearing clean and smooth with natural transitions between pixels. This suggests that the style transfer process has produced a visually pleasing result free from distracting graininess or artificial texture artifacts.

A low normalized value reveals significant noise contamination in the stylized image, manifesting as visible grain, speckling, or unnatural textural irregularities. Such artifacts can distract from the artistic quality of the result, create an unpolished appearance, and may indicate suboptimal optimization parameters or convergence issues during the style transfer process.

Unnatural Edges

Unnatural Edges measures the presence of artificial, jagged, or overly sharp edge patterns in the stylized image that do not correspond to natural structural boundaries. This metric identifies edges that appear abrupt, broken, or excessively emphasized, which often result fromoptimization artifacts, oversharpening, or the algorithm's struggle to smoothly integrate style patterns with content structures.

A high normalized value indicates that the stylized image contains natural, well-defined edges that follow coherent structural boundaries.

A low normalized value reveals the presence of unnatural edge artifacts in the stylized image. These may manifest as jagged outlines, broken contours, unnaturally sharp boundaries, or fragmented edges that disrupt the visual flow. Such artifacts can make the image appear unpolished, create visual confusion, and indicate that the style transfer process has introduced structural inconsistencies rather than gracefully integrating style with content.

Color Inconsistency

Color Inconsistency measures the uniformity and coherence of color distribution across the stylized image. It evaluates whether colors are applied consistently throughout the image or whether there are abrupt, unnatural shifts in hue, saturation, or brightness that disrupt visual harmony. This metric helps identify areas where the style transfer has failed to maintain consistent color application, resulting in patchy or fragmented color regions.

A high normalized value indicates that the stylized image exhibits smooth, consistent color transitions with uniform tonal quality across the entire image. Colors flow naturally from one region to another, creating a cohesive and visually pleasing result where the applied style appears evenly distributed without unnatural patches or discontinuities.

A low normalized value reveals significant color inconsistencies in the stylized image, manifesting as patchy color distribution, abrupt shifts in hue or saturation, or regions that appear disconnected from the overall color scheme. Such inconsistencies may arise when the style transfer algorithm struggles to balance conflicting color requirements between content and style, resulting in a fragmented appearance where different areas of the image seem to belong to different color palettes.

Unique Colors

Unique Colors measures the color depth and richness of the stylized image by quantifying the number of distinct RGB color values present. This metric serves as an indicator of posterization artifacts, where smooth gradients and subtle color variations become compressed into visible bands or blocks of uniform color. A diverse color palette typically indicates higher image quality and more faithful reproduction of the style's chromatic complexity.

A high normalized value indicates that the stylized image contains a rich and diverse color palette with smooth gradations and subtle variations. The image displays natural color transitions without visible banding, preserving the full chromatic complexity of the style while maintaining smooth tonal shifts that contribute to a professional, high-quality appearance.

A low normalized value reveals significant posterization in the stylized image, characterized by visible banding in gradients, flat color regions where natural variation should exist, and an overall simplified color palette. This degradation typicallyoccurs when the optimization process fails to preserve subtle color nuances or when compression artifacts reduce the effective color depth, resulting in an image that appears artificially simplified or lacking the chromatic richness of the intended style.

Coherence metrics

Texture Coherence

Texture Coherence (LBP - Local Binary Pattern) measures the preservation of fine-grained textural patterns from the original content image in the stylized result. Using Local Binary Pattern analysis, this metric evaluates how well surface textures, material appearances, and repetitive patterns—such as fabric weaves, skin pores, foliage details, or architectural surface details—are maintained after style transfer. It assesses whether the stylized image retains the textural character of the original content while incorporating the artistic style.

A high normalized value indicates that the stylized image successfully preserves the textural essence of the content image. Surface patterns, material appearances, and fine structural details remain recognizable and natural, with textures maintaining their original character even as colors and broader stylistic elements transform. This suggests that the style transfer has enhanced the image without destroying its inherent textural qualities.

A low normalized value reveals significant loss or distortion of the original texture patterns. The stylized image may exhibit flattened surfaces lacking material definition, merged or smeared textural details, or the introduction of unnatural repetitive patterns that do not correspond to either the original content or the intended style. Such texture degradation can make the image appear artificial, overly simplified, or visually confusing, as recognizable surface qualities essential to the content's identity become compromised.

Edge Coherence

Edge Coherence measures the preservation of structural boundaries and object outlines from the original content image in the stylized result. This metric evaluates how well the stylized image maintains the integrity of edges, contours, and major structural elements such as silhouettes, object boundaries, and the separation between distinct visual elements. It ensures that while artistic style is applied, the underlying structure of the scene remains intact and recognizable.

A high normalized value indicates that the stylized image successfully preserves the essential structural boundaries of the content image. Object outlines remain crisp and well-defined, major structural elements maintain their shape, and the spatial relationships between different components of the scene remain clearly distinguishable. The style transfer enhances the image artistically without compromising the structural clarity that makes the content recognizable.

A low normalized value reveals significant degradation or distortion of structural boundaries in the stylized image. This may manifest as blurred or softened edges where clear boundaries should exist, fragmented or broken object outlines, merging of distinct objects into ambiguous shapes, or complete loss of structural definition. Such degradation can make the content difficult to interpret, compromise object recognition, and indicate that the style transfer process has overwhelmed the essential structural information of the original image.

Aesthetic Quality metrics

Colorfulness

Colorfulness measures the vibrancy and richness of colors present in the stylized image. This metric evaluates the intensity and variety of chromatic content, assessing whether the image displays a lively, engaging color palette or appears muted, desaturated, or dull. It captures the overall chromatic energy of the result, which significantly influences its visual appeal and artistic impact.

A high normalized value indicates that the stylized image exhibits vibrant, rich colors with good saturation and chromatic variety. The image appears lively and engaging, with colors that feel intentional and artistically expressive. Such images capture attention and convey the intended stylistic energy effectively.

A low normalized value reveals that the stylized image lacks chromatic vibrancy, appearing muted, washed out, or predominantly grayscale. The colors may feel flat, uninteresting, or insufficiently expressive of the intended style. This can result from inadequate color transfer, overly aggressive smoothing, or style images that themselves lack strong chromatic character.

Contrast

Contrast measures the dynamic range between the brightest and darkest areas of the stylized image, evaluating the overall tonal separation and visual punch of the result. This metric assesses whether the image maintains good separation between light and shadow areas, creating depth, dimension, and visual interest, or whether it appears flat and lacking in tonal variation.

A high normalized value indicates that the stylized image exhibits strong tonal separation with well-defined highlights, midtones, and shadows. The image appears dynamic and dimensional, with visual depth that guides the viewer's attention and enhances the perception of form and structure.

A low normalized value reveals that the stylized image has poor tonal separation, appearing flat or washed out. The image may lack proper definition between elements, resulting in a visually uninteresting composition where details become difficult to distinguish due to insufficient variation in brightness levels.

Sharpness

Sharpness measures the clarity and definition of details in the stylized image, evaluating how well edges are rendered and fine details are preserved. This metric assesses whether the image appears crisp and well-defined or suffers from blurring, softness, or loss of fine structural information that reduces perceived quality.

A high normalized value indicates that the stylized image exhibits good clarity with well-defined details and crisp edges. Fine textures and subtle structural elements remain distinguishable, contributing to an overall impression of quality and precision. The image appears professionally rendered with appropriate focus on important visual elements.

A low normalized value reveals that the stylized image suffers from blurring or excessive softness, resulting in loss of fine detail definition. Edges may appear smeared, textures may become indistinct, and the overall image may seem out of focus or lacking in precision. This can result from over-smoothing during optimization or insufficient preservation of high-frequency information during the transfer process.

Composition

Composition measures how well the stylized image adheres to established visual design principles, specifically evaluating the placement of important visual elements relative to the rule of thirds. This metric assesses whether key features, edges, and points of interest align with the compositional guides that typically create visually pleasing and balanced arrangements.

A high normalized value indicates that the stylized image exhibits strong compositional structure, with important visual elements aligned with the rule of thirds intersections and guidelines. The arrangement of subjects, edges, and points of interest creates a balanced, visually pleasing layout that naturally guides the viewer's eye through the image.

A low normalized value reveals that the stylized image lacks strong compositional structure, with important visual elements poorly positioned relative to established compositional principles. The arrangement may feel unbalanced, awkward, or unintentional, potentially distracting from the artistic quality of the result regardless of how well style and content are otherwise preserved.

Overall score

The Overall Score provides a single comprehensive measure of stylized image quality by combining five distinct evaluation components into a weighted average. Each component captures a different dimension of quality, ensuring that the final score reflects both technical fidelity and artistic merit. The score ranges from 0 to 1, where higher values indicate superior overall quality. The Overall Score is calculated as follows:

Score =
0.40 × Perceptual Quality +
0.25 × Structural Coherence +
0.15 × Artifact Detection +
0.10 × Pixel Accuracy +
0.10 × Aesthetic Appeal

Each component is normalized to a 0–1 scale before weighting, with higher values always indicating better performance.